

SIM1002 Calculus I

Tutorial 5

Derivatives

Exercise 3.3

In following Exercises, find the first and second derivatives in the following questions.

1. $y = \frac{x^3}{3} + \frac{x^2}{2} + \frac{x}{4}$

2. $s = -2t^{-1} + \frac{4}{t^2}$

3. $r = \frac{12}{\theta} - \frac{4}{\theta^3} + \frac{1}{\theta^4}$

Find the derivatives of the functions.

4. $y = (2x + 3)(5x^2 - 4x)$

5. $y = (1 + x^2)\left(x^{\frac{3}{4}} - x^{-3}\right)$

6. $f(t) = \frac{t^2-1}{t^2+t-2}$

7. $y = \frac{(x+1)(x+2)}{(x-1)(x-2)}$

Find the derivatives of all orders of the functions.

8. $y = (4x^2 + 3)(2 - x)x$

9. $y = (x - 1)(x + 2)(x + 3)$

Find the first and second derivatives for Q10 - Q11.

10. $w = \left(\frac{1+3z}{3z}\right)(3 - z)$

11. $p = \frac{q^2+3}{(q-1)^3+(q+1)^3}$

12. Suppose u and v are differentiable functions of x and that
 $u(1) = 2, u'(1) = 0, v(1) = 5, v'(1) = -1.$

Find the values of the following derivatives at $x = 1$.

(a) $\frac{d}{dx}(uv)$ (b) $\frac{d}{dx}\left(\frac{u}{v}\right)$ (c) $\frac{d}{dx}\left(\frac{v}{u}\right)$ (d) $\frac{d}{dx}(7v - 2u)$

Slope and Tangents

13. The curves $y = x^2 + ax + b$ and $y = cx - x^2$ have a common tangent line at the point $(1,0)$. Find a, b and c .
14. The curve $y = ax^2 + bx + c$ passes through the point $(1,2)$ and is tangent to the line $y = x$ at the origin. Find a, b and c .

Exercises 3.5

Trigonometric functions

Find dy/dx in the following.

1. $y = \frac{3}{x} + 5 \sin x$
2. $y = x^2 \cot x - \frac{1}{x^2}$
3. $y = \frac{\cos x}{\sin^2 x}$
4. $y = \frac{\cos x}{x} + \frac{x}{\cos x}$
5. $y = (\sec x + \tan x)(\sec x - \tan x)$
6. $y = x^2 \cos x - 2x \sin x - 2 \cos x$

Find dp/dq .

7. $p = \frac{\tan q}{1 + \tan q}$
8. $p = \frac{3q + \tan q}{q \sec q}$